LITES (Library of IT and Engineering Studies): The Development of an Enhanced Library Management System for CEIT Reading Room

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Bachelor of Science in Information Technology

Mark John Ver N. Credito

Wilson Jhay-R E. Daganta

Angel Marie N. Dedase

Shawn Raven L. Ferrer

Darlene R. Soltes

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Mark John Ver N. Credito
Wilson Jhay-R E. Daganta
Angel Marie N. Dedase
Shawn Raven L. Ferrer
Darlene R. Soltes

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INTRODUCTION

An information system (IS) is technically defined as a set of interrelated components that retrieve, analyze, store, and disseminate information to enhance decision-making and control within an enterprise. These systems—comprised of hardware, software, and telecommunications networks—allow organizations to obtain data, make it relevant and facilitate its distribution. According to Bourgeois et al. (2019), the strategic implementation of information systems provides organizations with a competitive advantage by improving business processes and operational efficiency. In an academic setting, these systems are essential for managing the vast intellectual capital produced by students and faculty.

To be effective in specialized environments like academic libraries, information systems must serve as central repositories that organize, track, and secure high-value documents. Just as data management automation enables systems to process complex transactions accurately (Agustin, Syamra, & Putra, 2024), a dedicated management system in library setting ensures that institutional

knowledge is preserved and easily accessible. The success of such a system depends heavily on matching the technical expectations of system analysts with the specific needs of the users—in this case, the students and faculty of the College of Engineering and Information Technology(CEIT).

While management systems are often associated with financial institutions to minimize losses and organize borrower records (Arunkumar & Privietha, 2023), the same principles of centralization and data organization apply to the management of terminal research projects. A specialized Library Management System (LMS) for capstone and thesis projects acts as a digital archive that categorizes metadata, abstracts, and technical documentation. By moving away from manual logging and physical searching, a centralized digital framework leads to smoother operations and faster information retrieval.

The College of Engineering and Information Technology (CEIT) is one of the primary academic units of Cavite State University (CvSU) in the Philippines. Its main purpose is to provide high-quality technical education and foster innovative research among its faculty, staff, and students. As a hub for technological advancement, CEIT aims to provide superior academic support while promoting responsible research practices and intellectual property management among its community members. This aligns with the college's commitment to enhancing technological literacy and academic excellence, ensuring that the finished capstone and thesis projects—the culmination of its students' hard work—are preserved and made accessible through efficient, modern systems.

Hence, the researchers aim to develop an enhanced Library Management System (LMS) specifically for the College of Engineering and Information Technology (CEIT). This project focuses on digitalizing the archiving and retrieval of finished capstone and thesis projects to provide a more accessible, technologically advanced, and efficient system for the CEIT community.

Project Context

LITES: Library of IT and Engineering Studies for the College of Engineering and Information Technology (CEIT) Reading Room manages the specialized academic repository of the college, specifically focusing on archiving, monitoring, and retrieval of finished capstone, thesis, and design projects. The CEIT Reading Room serves as a vital resource for students and faculty, embodying the college's commitment to research excellence and technical innovation. Based on the current records and observations within the Reading Room at the Main Campus, it continues to serve a high volume of researchers, meeting a consistent threshold of over three hundred (300) visitors entering every semester. These visitors are managed by the Reading Room-in-Charge and library staff who are responsible for handling, maintaining, and organizing the collection of hardbound and softcopy research outputs from the previous programs under CEIT.

The researchers gathered data from the CEIT Reading Room indicating that while a previous system was delivered during the pandemic, it was never officially implemented or evaluated. Consequently, the Reading Room lacks a verified system to manage student attendance, book borrowing, and clearance signing. Every student under the CEIT programs is expected to submit their final requirements to the Reading Room, but the current manual workflow causes significant delays. As the college prepares for the renovation of the physical library, the staff expects a total digitalization of operations including a QR-based attendance system, automated borrowing and returning of the books, and a secure verification process for student clearances.

As the researchers conducted an interview with Mr. Ace Amiel E. Malicsi, the CEIT Reading Room-in-Charge and Research Coordinator, the problems stated and identified were: 1.) Time-Consuming Navigation and Search; 2.) Clearance

Verification and Accountability Issues; and 3.) Physical Space and Resource Limitations.

The first identified problem is the time-consuming manual navigation of research studies. According to the interview with Mr. Malicsi, it takes an excessive amount of time for students to find studies relevant to their own research because the current setup lacks a dynamic search engine. Students must manually browse through cabinets without the help of a digital catalog. Mr. Malicsi expressed the need for a system where students are given a unique ID and QR code that automatically detects their “Time In” and “Time Out” and allows them to search for studies using specific parameters such as program, keywords, and year. This would eliminate the need for manual handwriting in logbooks and drastically reduce the time spent on research retrieval.

The second identified problem is clearance verification and accountability. In the current process, students must obtain the signature of the Research Coordinator on their clearance by providing both a softcopy and hardbound copy of their thesis. However, problems occur when clearances are signed prematurely because printing offices often cannot produce hardbound copies immediately. Without physical evidence at the time of signing, some students receive signatures unnoticed without surrendering their hardbound books. To address this, Mr. Malicsi requires a system where a unique, non-decipherable code is generated once a student uploads their softcopy and abstract. This code, distinguishable only by Mr. Malicsi at a glance, will serve as a secure proof of submission that the student cannot forge or decipher.

The third identified problem is physical space and resource limitations. Due to the small capacity of the cabinets in the Reading Room, physical space for new hardbound copies has become very limited. Mr. Malicsi expressed that the room’s resources are stretched thin, highlighting the urgent need for a digitalized archiving process. The organization currently struggles to manage the bulk of physical

submissions, which increases risk of disorganized records. By digitalizing the abstract and metadata through LITES, and assigning each book a distinct QR code for online viewing, the Reading Room can optimize its limited space and provide better access to its 300+ seasonal visitors.

In addition, according to the Reading Room-in-Charge, Mr. Ace Amiel E. Malicsi, the root of these problems is that the CEIT Reading Room lacks a fully evaluated and enhanced integrated system to manage research legacies. These challenges highlight the urgent need for implementing LITES: Library of IT and Engineering Studies. The Development of this system will enhance accessibility, embrace advanced QR technology, and simplify computerized workflows to align with CEIT's vision of academic excellence and modernized library operations.

Objectives of the Study

The main objective of the study is to design and develop an enhanced Library Management System (LMS) for the College of Engineering and Information Technology (CEIT) that specifically manages and archives finished capstone and thesis projects.

Specifically, the study aims to:

1. Identify and analyze the existing problems in the current manual or existing archiving and retrieval processes of the CEIT Library;
2. Design a Library Management System (LMS) that is capable of:
    - Digitizing the submission, recording and organization of finished capstone and thesis projects;
    - Providing an efficient search and filter feature (e.g., by title, author, year, or technology used) for easier research retrieval;
    - Ensuring a user-friendly interface for both library staff (administrator) and student (researchers);

- Securing digital copies (PDFs) of research projects to prevent unauthorized access or copying;

3. Develop the System Using PHP, Javascript, and React, with MySQL (XAMPP) for the database management;
4. Test the system's functionality, user experience, usability, and responsiveness across web-based platforms;
5. Evaluate the system based on the ISO 25010 software quality standards (Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Probability); and
6. Prepare an implementation plan for deployment and maintenance of the enhanced Library Management System within the CEIT Library.

Purpose and Description

The Primary objective of this project is to develop an Enhanced Library Management System (LMS) for the College of Engineering and Information Technology (CEIT) to meet the specific requirements of archiving and managing finished capstone and thesis projects. With a focus on increasing operational effectiveness and researcher satisfaction, the system aims to simplify and optimize the submission, categorization, and retrieval procedure of academic research.

The Enhanced Library Management System will apply these capabilities:

1. Search and Retrieval Efficiency: Providing an intuitive and user-friendly interface for students and faculty to easily search for past research projects. This transparency helps researchers stay informed about existing studies and avoid duplication of topics.
2. Automated Archiving Workflow: Organizing the thesis submission workflow to improve efficiency and reduce manual processing time. This ensures that

metadata—such as titles, authors, and technical abstracts—are captured accurately and quickly.

3. Digital Content Security: Optimizing the access process to ensure that digital copies of projects are protected, allowing only authorized users to view or read the full content within the library's premises.
4. Comprehensive Research History: Offering a digital catalog functionality where users can view the repository's history, including details such as the year of publication, technical stacks used, and the specific engineering or IT department the project belongs to.
5. Project Category Filtering Tool: The system includes a filtering tool that allows researchers to browse projects based on different categories, such as "Internet of Things," "Mobile Applications," or "Machine Learning." This empowers students to make informed decisions when selecting their own research technologies.
6. Upholding ISO 25010 Standards: Ensuring the security and reliability of the system, thereby safeguarding the intellectual property of the university and the personal data of the student authors.

The Enhanced Library Management System aims to promote broader access to institutional knowledge while supporting sustainable development objectives linked to digital transformation and academic resource efficiency. The CEIT Library faces challenges in manual record-keeping and physical space limitations for hardbound copies. This study will help the college establish an easier process by providing an effective, centralized system for managing its research legacy.

Time and Place of the Study

The research begins upon the approval of the thesis proposal and concludes with the presentation of the system results and final defense during the Second Semester of Academic Year 2025-2026.

The data gathering, system requirement analysis, and initial user testing will take place at the College of Engineering and Information Technology (CEIT), located within the Cavite State University (CvSU) Main Campus in Indang, Cavite. System development and programming will be conducted by the researchers throughout the designated period, while final system evaluation and validation will be held on the college premises to ensure the software meets the specific needs of the CEIT students and library staff.

Scope and Limitation of the Study

The study titled “Development of an Enhanced Library Management System for the College of Engineering and Information Technology (CEIT)” aims to automate and optimize the archiving and retrieval processes of finished capstone and thesis projects. The system focuses on improving the accessibility of research metadata while ensuring the security of intellectual property and digital records. The primary users of the system will be the CEIT Library Staff (Admin) and the CEIT students and faculty. The System will be implemented specifically within the CEIT Library at Cavite State University (CvSU) Main Campus.

The system includes various modules designed to create a cohesive digital repository:

Account Management Module

This module manages user access to the repository. It provides secure login forms for students and staff. For administrators (Librarians), the module provides tools to manage user permissions, oversee account activity and verify student credentials. Students can create and manage their profiles, allowing them to save “bookmarked” researched titles or track the status of their submitted electronic copies.

Thesis Archiving and Repository Module

This serves as the central tool for managing the collection of finished projects. It allows librarians to record new entries, upload digital abstracts, and meticulously categorize projects by year, course (e.g., BSIT, BSCS, BSCE), and technical specialization. This module ensures that all academic records are stored in a structured database rather than physical logbooks.

Search and Retrieval Module (Web-Based)

This module provides a user-friendly interface for researchers to navigate the library's collection. Users can search for projects using keywords, titles, or author names. An advanced filtering feature allows researchers to sort projects by technical stack (e.g., "React," "IoT," "Arduino") or academic year, providing instant feedback on the availability of specific research topics.

Project Category and Metadata Module

This module allows administrators to organize projects into specific technological domains. It enables the system to generate analytics on research trends within the college. For students, it serves as a guide to see which areas of study have been extensively covered and which are currently emerging, helping them make informed decisions for their own future capstone projects.

Submission Tracking and History Module

Both students and administrators can access a comprehensive record of project submissions. For students, this includes the status of their hardbound and softcopy submissions (Pending, Verified, or Archived). On the admin side, this provides a log of all administrative actions, such as file uploads, record modifications, and system audits, ensuring the integrity of the library's data.

Content Management Module

This module handles the organization and distribution of digital content across the platform. Administrators can easily update library policies, upload

new submission guidelines, and manage digital forms. It ensures that all information available to students regarding thesis requirements remains accurate, relevant, and compliant with current CEIT standards.

Report Generation Module

This module collects and analyzes data regarding library usage and the research collection. It allows administrators to generate reports such as the total number of projects per department, most searched keywords, and delinquency reports for students who have not yet completed their submission requirements. Data visualization, including charts and tables, is provided for the library's annual reports, with options to export these summaries into PDF format for official documentation.

Limitation

The system is specifically designed for the CEIT Library's collection of finished capstone and thesis projects and does not include the general circulation of textbooks or other library materials. Access to full digital manuscripts may be restricted to authorized users within the campus network to comply with intellectual property policies.

Conceptual Framework of the Study

In this section, the conceptual framework is presented, illustrating the inputs, iterative development process using modern technical stacks, and the intended impact of LITES: Library of IT and Engineering Studies for the CEIT Reading Room.

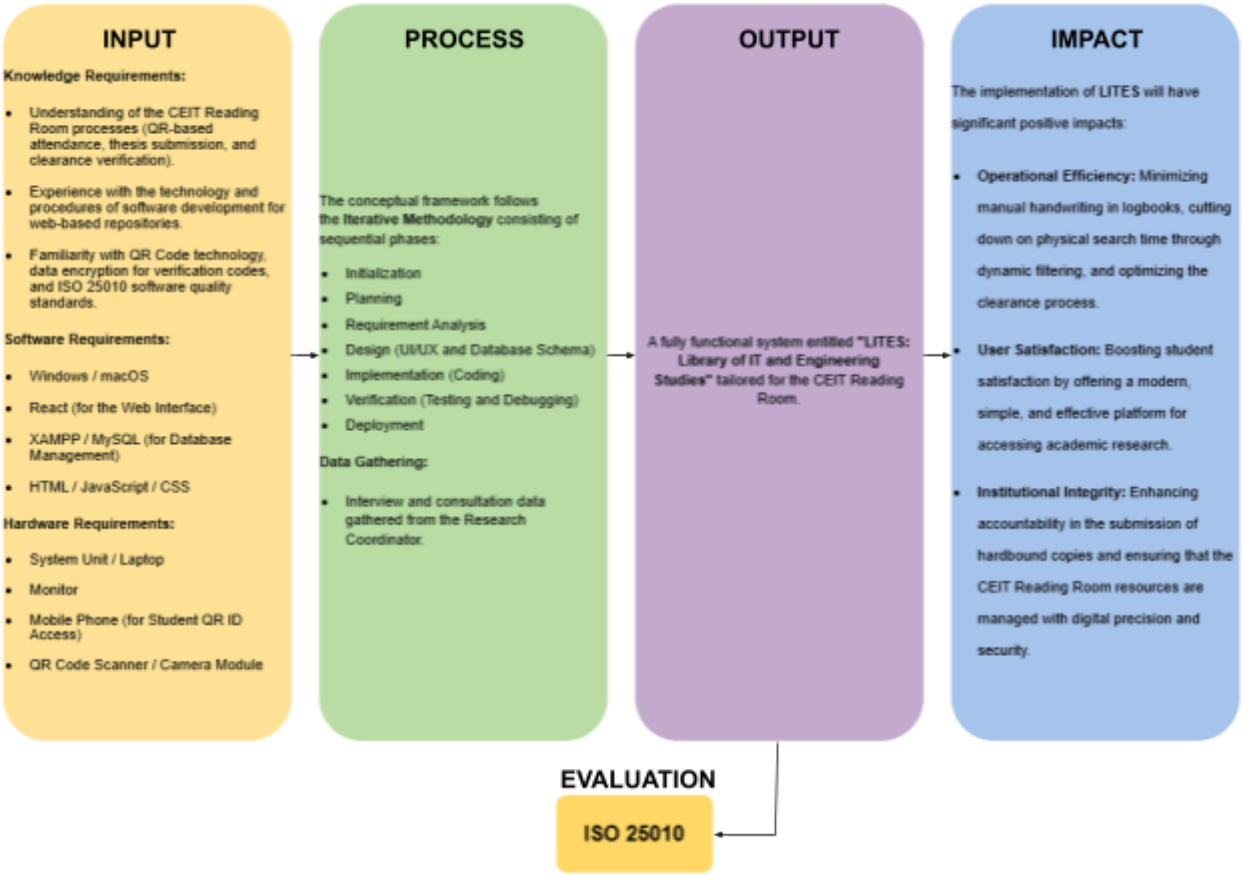

Figure 1. Conceptual framework of LITES: Library of IT and Engineering Studies

The conceptual framework for LITES is designed to address the specific challenges faced by the CEIT Reading Room, such as inefficient navigation and the manual clearance process. This system aims to digitalize library operations while adhering to international software quality standards like ISO 25010.

The first step involves understanding the Reading Room's current operations and the needs of its visitors. The conceptual framework outlines the inputs required for development, which include technical expertise in web development, specific software and hardware requirements, and an understanding of library archiving protocols.

Following the Iterative Development Methodology, the framework outlines a step-by-step process for developing the system. This involves planning, designing, coding, and testing phases to ensure that the system is "enhanced" based on the

previous student projects. The anticipated output is a fully functional repository tailored for CEIT. The system is expected to improve student research efficiency, optimize archiving procedures, and ensure that the Research Coordinator has a secure way to verify submissions. By leveraging QR Code technology and a systematic development process, LITES will promote digital transformation and academic resource efficiency within the CEIT community.

Definition of Terms

Administrator. The Individual, specifically the CEIT Reading Room-in-Charge, responsible for managing the user accounts, approving thesis uploads, and overseeing the overall digital repository.

Archiving. The process of collecting, categorizing, and securely storing finished capstone and thesis projects in a digital format to ensure long-term preservation and accessibility.

Borrowing. The process of temporarily checking out a physical hardbound thesis from the Reading Room, which is tracked in the LITES system via QR Code Scanning.

CEIT Reading Room. The specific academic facility within the College of Engineering and Information Technology at Cavite State University for which the LITES system is developed.

Clearance Verification Code. A unique, non-decipherable alphanumeric code generated by LITES once a student successfully uploads their softcopy and abstract, used by the Research Coordinator as proof of submission.

Compliance. Adherence to the academic policies of the university and the technical standards of software development to ensure the system meets institutional requirements.

Digitalization. The conversion of manual library processes, such as logbook entries and physical document tracking, into automated digital workflows.

Dynamic Searching. A system feature that allows researchers to filter through the thesis repository using multiple parameters such as program, year, keyword, and technical stack.

History. A comprehensive digital record of past activities within the system, including a student's borrowing history, previous research submissions, and administrative logs.

Implementation. The phase initiated after LITES has been tested and accepted by the Reading Room staff, where the system is installed and integrated into daily library operations.

Initialization. The starting phase of the system development where the environment is set up and the initial database structures are created.

ISO 25010. The international standard used to evaluate the quality of the LITES software, focusing on attributes such as functional suitability, security, and usability.

LITES (Library of IT and Engineering Studies). The enhanced library management system developed to digitalize and optimize the operations of the CEIT Reading Room.

Metadata. Specific data that describes the research projects, such as the title, authors, abstract, keywords, and the technology used (technical stack).

MySQL. The open-source Relational Database Management System (RDBMS) used by LITES to store and manage student profiles, research metadata, and transaction logs.

QR Code (Quick Response Code). A machine-readable code consisting of an array of black and white squares, used in LITES for automated student attendance (Time-In/Time-Out) and for identifying physical thesis books.

Reading Room-in-Charge. The primary staff member responsible for the operations of the CEIT Reading Room and the main user of the LITES administrative module.

Report. A structured document generated by the system that presents data on library usage, the number of archived projects, and lists of student submissions or delinquencies.

Research Coordinator. The faculty member responsible for signing student clearances, who utilizes the LITES verification code to ensure all submission requirements are met.

Softcopy. The digital version of the finished capstone or thesis project (usually in PDF format) that students upload to the LITES repository.

Staff. The employees assigned to the CEIT Reading Room who assist in the daily operations of borrowing, returning, and archiving research materials.

Status. The current state of a student's submission or a book’s availability, such as "Pending," "Archived," or "Borrowed."

System. The integrated combination of the web interface, QR technology, and the MySQL database designed to manage the CEIT Reading Room's research legacy.

Technical Stack. The set of technologies (e.g., React, JavaScript, MySQL) used to build and run the LITES application.

Tracking. The automated monitoring of student attendance and the movement of thesis books within the Reading Room through the use of QR scanners.

REVIEW OF RELATED LITERATURE

This section explores existing related literature and studies about Loan Management Systems.

System Technical Background

The development of LITES: Library of IT and Engineering Studies utilizes a modern web-based stack to ensure high performance, security, and a responsive user interface. The system leverages React for the front-end architecture, MySQL for robust data management, and XAMPP as the local server solution. Unlike the traditional manual systems, LITES integrates QR Code Technology to automate attendance and book tracking, necessitating a framework that can handle real-time data updates without page reloads.

React. According to Rawat and Mahajan (2020), ReactJS is an open-source library utilized for building user interfaces explicitly for single-page applications. In the context of LITES, React allows researchers to use dynamic filtering—such as searching by technical stack or year—without needing to refresh the page. This is achieved through a "clever diffing mechanism" that updates only the necessary data in the DOM. As noted by Saks (2019), React simplifies the process of updating the view when data changes by breaking the UI into smaller, reusable components, which is essential for managing the complex metadata of engineering and IT research projects.

Database Management Systems. According to Rawat, Purnama, and Mulyati (2021), a Database Management System (DBMS) is vital as the volume of data grows, facilitating the categorization and handling of data into necessary information. For the CEIT Reading Room, which manages hundreds of thesis records, a DBMS is crucial for request management and user administration. User management oversees authentication

to safeguard research data from unauthorized access, while request management allows the system to determine which studies are frequently sought by visitors.

QR Code Technology. Quick Response (QR) codes are two-dimensional matrix barcodes that offer transformative opportunities for academic libraries to enhance user experience and streamline operations. According to Pareek (2024), QR codes bridge the gap between physical resources and digital data, providing a fast and simple method for users to access online databases and digital content. In a library setting, QR codes facilitate "contactless" services, such as touchless borrowing and automated attendance tracking, which address modern institutional needs for efficiency and resource management.

ISO 25010 Software Quality Standards. ISO/IEC 25010 is an international standard used to evaluate the quality of software products across eight main dimensions: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. According to recent academic evaluations, integrating this standard into a research management system ensures that the technical development is based on measurable and validated parameters. This model is essential for verifying that a digital repository successfully addresses institutional requirements while providing a secure and dependable platform for stakeholders.

Digital Repositories and Dynamic Search. Modern digital libraries serve as transformative tools that reshape how academic research is managed and stored, providing seamless access to information. According to Gong (2025), advanced search features such as filtering by educational background or research preference allow libraries to provide targeted offerings that better fulfill users' knowledge needs. Furthermore, intelligent search engines and metadata-driven filtering enable researchers to navigate vast databases easily, preventing the duplication of research topics and fostering a more interactive learning environment.

MySQL. As discussed by Shuvo, Tuhin, and Parvez (2023), MySQL is based on Structured Query Language (SQL), used for adding, removing, and modifying information. In LITES, MySQL stores critical research metadata, student profiles, and the unique QR-linked transaction logs. Standard SQL commands ensure that the Reading Room staff can generate accurate reports on research trends and submission statuses.

XAMPP. According to Shuvo, Tuhin, and Parvez (2023), XAMPP is a free and open-source cross-platform web server solution stack package consisting mainly of the Apache HTTP Server, MariaDB (or MySQL) database, and interpreters for scripts. In the development of LITES, XAMPP provides the essential local server environment needed to host the system's backend logic and manage the repository's database before full deployment.

In terms of networking, the system functions primarily in a localhost or local area network (LAN) setting, enabling seamless communication between the LITES web interface, the MySQL database, and the QR scanning modules. The use of a Relational Database Management System (RDBMS) like MySQL provides the necessary performance and scalability to handle the growing volume of engineering and IT research metadata. For LITES, a Model-View-Controller (MVC) architecture is considered during development to separate the data (thesis records), the logic (borrowing/attendance rules), and the interface (student and admin portals), ensuring the system is maintainable and adaptable for future enhancements.

LITES is constructed with effectiveness, security, and scalability in mind by utilizing the React development environment, the JavaScript framework, and the XAMPP server. This configuration ensures that the system satisfies the needs of the CEIT Reading Room and the Research Coordinator for a dependable digital archive and automated monitoring platform.

Mobile and Web Access Strategies. According to Zopp (2020), mobile accessibility can be achieved through native, hybrid, or cross-platform approaches. For LITES, providing students with the ability to scan book QR codes and view abstracts via their smartphones is a primary requirement. While native development offers the most efficient app lifecycle, cross-platform approaches—like those utilized in React-based frameworks—provide a single codebase that can operate efficiently across different devices.

Unlike traditional Progressive Web Applications (PWAs) which may suffer from limited access to mobile hardware, the LITES framework aims to leverage responsive web design to ensure that students can use their mobile cameras to interact with the Reading Room's physical resources. This ensures that the system remains accessible to the 300+ students of the college, regardless of whether they are accessing the repository via a desktop terminal in the Reading Room or through their personal mobile devices.

Related Literature

This section explores the conceptual and theoretical foundations of modern library operations, digitalized archiving, and institutional accountability. These literatures provide the framework for the functionalities integrated into LITES.

Concept of Library Management and Automation. According to digitalcommons.unl.edu (2020), library automation is the application of ICT to library operations to provide a more efficient way of managing transactions such as issuing and returning books. Automated systems allow librarians to maintain computerized databases instead of manual logbooks, which bridges research gaps and addresses operational "pain points." Modern automation is no longer just about tracking books but has evolved

into transforming library services through digital platforms that allow for better data management and accessibility.

Academic Repositories and Research Preservation. An institutional repository serves as a digital archive for the intellectual output of an academic institution. According to Crow (2002), these repositories are a set of services that a university offers to its community for the management and dissemination of digital materials created by the institution and its community members. In an engineering and IT context, these repositories are essential for preserving technical legacy, allowing students to examine past technical methodologies and technical stacks, which prevents the duplication of research topics.

QR Technology in Institutional Monitoring. Quick Response (QR) codes have become a standard tool for bridging physical and digital environments. According to Pareek (2024), QR codes provide a user-friendly and fast method for spreading data, allowing users to access online databases by scanning a physical resource. In institutional monitoring, literature suggests that QR code-based attendance systems are more secure and cost-effective than traditional methods because they utilize existing mobile hardware to record "Time-In" and "Time-Out" data in real-time, significantly reducing the administrative burden.

Electronic Clearance and Submission Policies. The transition from paper-based to electronic clearance systems is a critical concept in modern school administration. According to Mangubat et al. (2023), electronic clearance systems allow for a paperless approach where students can monitor their status remotely. The literature highlights that digitalizing the clearance process ensures high data integrity and provides real-time updates for signatories. This is particularly relevant in verifying that physical requirements, such as hardbound copies, are officially recorded in the system before a clearance is granted, thereby closing the accountability gap found in manual signatures.

User Interface and Experience (UI/UX) in Academic Systems. The success of a digital system relies heavily on matching the expectations of its users. According to Agustin, Syamra, & Putra (2024), the development of information systems must prioritize the interaction between the system and the user to ensure usability. For academic repositories, a user-friendly interface is necessary to allow students to navigate complex research metadata easily. Effective UI/UX design in libraries ensures that features like dynamic searching and filtering are intuitive, encouraging the frequent use of the digital archive.

Related Studies

The section below offers an overview of relevant studies in the field and presents the review related studies from local and foreign sources. This chapter aims to provide a comprehensive review of external literature relevant to the researchers' chosen topic, along with a review of similar studies conducted previously.

Modern Library Management System. The findings from Bhati, Komal, Prince & Saxena (2024) describe a Library Management System (LMS) designed as a web-based platform that offers centralized control over resources and enables real-time updates. The study emphasizes that automated validation and modular design significantly reduce manual overhead and minimize human errors. This research identifies critical limitations in conventional systems, such as the lack of automated reminders and smart identification. While this study proposes future readiness through smart card-based identification, LITES enhances this concept by utilizing QR Code technology for immediate, cost-effective student identification and attendance tracking in the CEIT Reading Room.

Evaluating User Satisfaction with Digital Library Services. According to Abu, Azib, Rafie & Tokiran (2025), the transformation of libraries from physical spaces to dynamic online hubs has reshaped information management, particularly in higher education. Their research highlights that digital libraries are essential repositories that ensure both accessibility and quality of scholarly content. This study focuses on identifying factors that influence student satisfaction to create more user-centered platforms. This is highly relevant to LITES, as the researchers aim to address the UI/UX failures of the previous system to ensure that the CEIT Reading Room provides a high-level, satisfying digital experience for its 300+ seasonal visitors.

Website-Based Library Information System for Academic Libraries. According to Albab, Bakhar, and Sungkar (2025), a web-based information system was designed to resolve issues of error-prone manual recording and slow data retrieval. Developed using the Waterfall method with PHP and MySQL, the system includes features for borrowing, returning, and member management. While their system successfully improved the speed of access to library information, LITES advances this architecture by specifically targeting the "accountability gap" in research archiving. Unlike general library systems, LITES integrates a specialized Research Coordinator verification code to ensure that digital and physical thesis requirements are met before clearance is granted.

Student E-Library Card Automated Attendance Using QR Code Scanning. According to Scribd (2025), this study at Romblon State University aimed to improve the accuracy and efficiency of attendance tracking by replacing traditional manual logbooks with a QR code scanning system. The researchers emphasized that computerizing library attendance allows data to be transformed into a centralized database for real-time reporting and later analysis. Similar to LITES, this system

reduces the administrative burden on library staff by avoiding the maintenance of physical registry books and providing accurate visitor statistics.

Design and Develop a Thesis and Capstone Project Manuscript Archiving Management System. According to Perante et al. (2024), this study for Eastern Visayas State University - Ormoc Campus addressed the challenges of traditional paper-based archives where manuscripts were kept in physical designated areas. The researchers developed a web-based digital platform utilizing an Agile and Rapid Application Development (RAD) approach to ensure an adaptable and iterative development process. The study highlights that existing archival systems often lack essential features for managing complex thesis metadata, a problem LITES aims to solve through its specialized technical stack filtering and secure clearance verification.

Table 1.
Comparison of Review Related Studies.

Aspects	RS 1	RS 2	RS 3	Research
Title and Authors	Modern Library Management System. Bhati, Komal, Prince & Saxena (2024)	Website-based Library Information System. Albab, Bakhar & Sungkar (2025)	Thesis and Capstone Manuscript Archiving System. Perante, et al. (2024)	Development of Loan Management System for Cavite State University Multipurpose Cooperative (2024)
Purpose	To provide a web-based framework for resource automation and centralized user oversight.	To enhance efficiency and accuracy in academic library management using web technologies.	To develop a digital platform for the archiving and preservation of student research manuscripts	To digitalize the CEIT Reading Room operations, focusing on research archiving, QR attendance, and secure clearance.
Research Methodology	Modular Design and Automated Validation approach.	Waterfall Method for structured system development.	Agile and Rapid Application Development (RAD) approach.	Iterative Process Methodology (Requirement, Design, Implementation, Testing, and Deployment).
Programming Languages/ Technologies Used	Web-based Framework (Unspecified)	PHP and MySQL	Web-based System, Digital Archiving Tools	JavaScript, React, XAMPP, MySQL, HTML, QR Technology

Table 1. Continued…

Features/ Functionality	Centralized resource control, real-time updates, and automated validation.	Book borrowing/returning records and customized admin dashboards.	Digital Manuscript archiving, metadata management, and secure storage.	QR-based Time-in/out, Dynamic Research Search, Unique QR for Books, and Secure Coordinator Codes.
Implications	Reducing overhead of manual operations and minimizing human errors.	Improved speed and ease of access to library information for students.	Preservation of institutional research legacy and improved digital accessibility.	Smooth research retrieval, automated monitoring, and gap-free clearance accountability.
Conclusion	Establishes a robust foundation for secure, user-centric library automation.	Successfully improved library efficiency and accuracy in record-keeping.	Addresses the lack of features in traditional archives for managing complex metadata.	LITES will modernize the CEIT Reading Room by embracing QR technology and securing the submission workflow.

Synthesis

The development of LITES: Library of IT and Engineering Studies for the CEIT Reading Room leverages a modern web-based framework known for its high performance and scalability. This system runs in a robust development environment utilizing MySQL for database management and XAMPP for local server solutions. The front-end architecture is developed using React, ensuring a responsive and interactive user experience for both student researchers and library administrators.

A standard desktop or laptop computer capable of running modern development applications and a high-resolution display monitor are the primary hardware requirements for the system's construction. The software stack, consisting of React, XAMPP, and MySQL, serves as the cornerstone for building this digital repository. As explained by Rawat and Mahajan (2020), utilizing React allows developers to manage complex data manipulation and metadata updates without requiring full-page refreshes, which is essential for the dynamic searching and filtering of thesis abstracts within a single-page application (SPA).

While mobile accessibility is often associated with native applications, the cross-platform capabilities of the selected technical stack ensure that LITES can optimize its design for various devices. By utilizing QR Code technology for automated attendance and book tracking, the system guarantees dependable and effective performance across different smartphone setups. This aligns with the findings of Sardjono (2023) regarding the optimization of mobile-centric interfaces for diverse user configurations.

According to Rawat, Purnama, and Mulyati (2021), the MySQL Database Management System (DBMS) is essential for effectively managing and classifying enormous amounts of academic data. In the context of the CEIT Reading Room,

MySQL’s capabilities enable smooth data transformation and efficient querying of research titles, technical stacks, and student profiles. Furthermore, Shuvo, Tuhin, and Parvez (2023) explain that XAMPP provides a complete web server solution that facilitates the local hosting of the Apache server and the management of the MariaDB/MySQL database.

The main operating environment for LITES is a local network or localhost setting, which facilitates smooth communication between the web application layers and the database components. With a focus on Model-View-Controller (MVC) architecture, the system is built for scalability, reliability, and security. By utilizing this technological stack and adhering to the ISO 25010 quality standards, LITES seeks to satisfy the CEIT Reading Room's requirements for a dependable digital archive and a secure clearance verification process. This ensures both student satisfaction and operational excellence, ultimately bridging the accountability gap in the college's research management.

METHODOLOGY

This chapter reveals the research methodology, documentation requirements—including functional and non-functional specifications—system implementation, and the specific features of LITES.

Requirements Analysis

The development of LITES: Library of IT and Engineering Studies for the CEIT Reading Room requires a thorough understanding of the current academic archiving procedures to meet the expectations and needs of the college, located in Indang, Cavite. Building on insights gained from extensive interviews with stakeholders, specifically Mr. Ace Amiel E. Malicsi, the CEIT Reading Room-in-Charge, the study aims to define user requirements comprehensively. The researchers' goal is to actively engage with the Reading Room staff and the Research Coordinator to understand their specific needs and preferences to implement the system efficiently.

In addition, the analysis concentrates on determining the functional needs for important system features, including QR-based attendance tracking, dynamic thesis search, and digitalized clearance verification. Creating user-friendly interfaces (UI/UX) is highly prioritized to ensure accessibility for both students and employees, specifically addressing the feedback that previous systems lacked usability. Different roles and permissions will be created to meet the various needs of stakeholders, such as the Administrator (Reading Room-in-Charge), the Signatory (Research Coordinator), and the User (CEIT Student).

In order to meet the concerns expressed by the CEIT administration, security requirements are addressed with an emphasis on protecting the intellectual property of the research manuscripts. This includes defining strong data security controls and

information security procedures, ensuring that the unique verification code for clearances remains non-decipherable by students and that sensitive user data is handled with integrity.

In order to establish the necessary data items and linkages between entities for effective data management, database requirements are carefully examined. The technical requirements include the use of tools such as XAMPP for local server management, React for web-based front-end development, and MySQL for database management. These tools are selected to ensure that LITES can handle high-traffic periods where over 300 students access the repository within a semester. All users within the CEIT Reading Room's facilities are guaranteed to have access to designated computer terminals and the university's network infrastructure.

Strong technical support systems, comprehensive documentation, and efficient user training are essential for the success of LITES. These components enhance system utility and user satisfaction. In order to meet CEIT's commitment to academic excellence and research transparency, additional capabilities—including user activity logs, automated research trend reports, and QR-enabled book borrowing—will be implemented to ensure smooth and modernized operations.

Requirement Documentation

The following table consists of main features, functional and non-functional requirements of the Library Management System for CEIT Reading Room.

Table 2. Features of LITES

MAIN FEATURES	DETAILED FEATURES
User Login Feature	- Provides security for registered CEIT students and staff to access the repository. - Authentication is restricted to authorized CEIT accounts to ensure institutional data privacy.

User Profile Feature	-	Allows students to update personal details, manage their unique Library ID, and view their research submission history.
QR-Based Attendance	-	Automatically records “Time-In” and “Time-Out” upon scanning the student’s unique QR ID.
	-	Eliminates manual logbook entries and provides real-time visitor logs for the Reading Room.
Dynamic Research Search	-	Enables users to search and filter the archive by program (e.g., BSIT, BSCS, BSEE, etc), keywords, publication year, and specific technical stacks.
Digital Thesis/ Capstone Archiving	-	Allows students to upload the softcopy (PDF) of their finished capstone or thesis and enter relevant metadata (Abstract, Authors, Advisers).
QR Book Identification	-	Generates a distinct QR code for every approved thesis book.
	-	Students can scan the book’s QR code to instantly view the abstract and metadata on their mobile devices.
Submission Status Tracking	-	Enable students to track the status of their thesis submission (e.g., Pending, Under Review, or Archived) from the initial upload to final acceptance.
Borrowing & Return Feature	-	Automates the tracking of physical hardbound copies using QR scanners to record the date borrowed and the expected return date.
Secure Clearance Verification	-	Generates a unique, non-decipherable verification code for the Research Coordinator once the student completes the digital upload requirement.
	-	Serves as a secure proof of submission for the signing of the student’s clearance.

Automated Report Generation	-	Allows administrators to generate structured reports on the number of projects archived, visitor statistics, and lists of delinquent submissions.
In-App & Email Notifications	-	Sends automated updates to students regarding their submission approval status, overdue borrowed books, or document requests from the CEIT Reading Room-in-Charge.
Content Management	-	Allows the Reading Room-in-Charge to manage research categories, update submission guidelines, and oversee the digital repository's growth.

Functional Requirements

Table 3. Functional Requirements of LITES

MAIN FUNCTION(S)	DETAILED REQUIREMENTS / DESCRIPTION(S)
Research Archiving Workflow	1. Develop an intuitive and user-friendly interface for students to initiate the softcopy upload process. 2. Incorporate an approval workflow for the administrator to evaluate and accept uploaded manuscripts and metadata objectively. 3. The system will automate the archiving of research titles faster and smoothly than manual logging.
Data Security & Intellectual Property	1. Mandate user authentication (using CEIT credentials) for accessing the system. 2. Implement access control based on user roles (Student, Admin) to maintain data confidentiality. 3. Employ encryption techniques to safeguard digital manuscripts (PDFs) against unauthorized access, copying, or downloads outside the campus network.

Research Tracking & Profile Management	1. Allow students to view their submission history and the status of their hardbound clearance (e.g., Pending, Approved, Archived). 2. Allow students to update their profiles securely through the system. 3. Allow students to view their automated attendance records and borrowing history.
Repository/Report Review Management	1. Generated detailed academic reports related to the thesis portfolio, including number of submissions per program, technical stack used, and year of publication. 2. Implement tools for tracking “delinquent” submissions to monitor and manage potential clearance issues or missing documents. 3. Display key indicators such as research distribution, visitor frequency, and portfolio growth over time.
Automated QR Monitoring	1. The system must utilize a scanner module to automatically detect and record student Time-In/Time using their unique QR ID. 2. The system must generate a non-decipherable verification code for the research coordinator to authenticate final thesis submission before signing physical clearances.

Non-functional Requirements

Table 4. Non-Functional Requirements of LITES

NON-FUNCTIONAL REQUIREMENTS	DETAILED REQUIREMENTS / DESCRIPTION(S)
1. Performance	1. The system should be able to handle at least 50 simultaneous users (student/staff) without any noticeable decrease in performance during peak hours. 2. The response time for search queries (dynamic filtering) should not exceed 3 seconds under normal system load.

	3. The system must have at least 99.9% minimum uptime during academic operating hours.
2. Scalability	1. Accommodate an increase in the number of users (CEIT student population growth) and thesis/capstone project submissions annually. 2. It should be able to handle at least 20% increase in user base and data volume annually without requiring significant architecture changes.
3. Reliability	1. The system should have backup and recovery features/mechanisms to ensure data integrity and availability of research manuscripts. 2. The system should be able to survive crashes and ensure quick restoration without significant data loss of submission logs or student profiles.
4. Security	1. Use appropriate cryptographic algorithms for encryption of sensitive data (student information and the Coordinator Verification Code) exchanged between client and server. 2. The system should be developed according to the data privacy and security policies of the university regarding intellectual property protection. 3. Security audits and penetration testing should be conducted regularly to identify any weaknesses.
5. Usability	1. The system should have a user-friendly interface that aligns with modern UI/UX standards, with intuitive navigation and clear instructions. 2. It should be accessible from different devices and screen sizes (responsive web design) for scanning QR codes and viewing abstracts.
6. Compatibility	1. The system should be compatible with commonly used web browsers such as Chrome, Edge, Safari, and Firefox. 2. It should be compatible with standard operating systems (Windows, macOS, Android, iOS) used by students for accessing the web portal.
7. Maintainability	1. The system should be modular and well-documented (source code comments, technical specs) to promote ease in its maintenance and future enhancements. 2. Provide a mechanism for easy deployment of updates and patches without disrupting academic services during the semester.
8. Auditability	1. The system should maintain detailed logs of all user activities, including login attempts, PDF

	submissions, attendance (Time-In/Out), and librarian approvals. 2. Logs should be stored securely and accessible only to authorized personnel (CEIT Admin/IT Department) for auditing purposes.
9. Regulatory Compliance	1. The system should comply with all relevant laws and regulations governing academic records management and data protection (e.g., the Philippine Data Privacy Act)

Context Diagram

The context diagram reflects the interaction between the primary actors of the system: the CEIT Students, the Reading Room-in-Charge (Admin), and the Research Coordinator (Signatory). In this diagram, the Reading Room-in-Charge is primarily responsible for managing the digital repository, approving thesis uploads, and overseeing user accounts and borrowing records. The Students interact with the system by scanning QR codes for attendance, searching for research manuscripts, and uploading their finished capstone or thesis files. Furthermore, the Research Coordinator acts as a specialized actor who interacts with the system to verify the student submissions through a secure, non-decipherable code before signing terminal clearances.

The LITES (Library of IT and Engineering Studies) context diagram showcases its centralized interaction with the CEIT community and the internal research database. Students can view their attendance logs, track their submission status, and scan book QR codes to view metadata. Administrators manage the overall flow of library resources, generate research trend reports, and validate the authenticity of submitted softcopies. By digitalizing these interactions, the system

effectively replaces the manual logbook and paper-based tracking methods with a secure and automated digital framework.

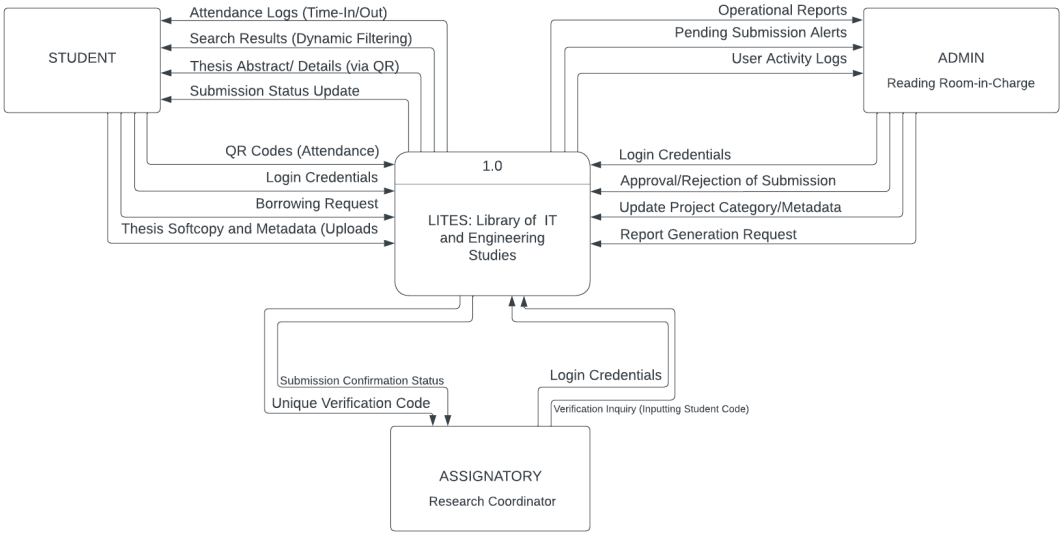

Figure 2. Context Diagram of LITES: Library of IT and Engineering Studies for CEIT Reading Room

Use Case Diagram

Figure 2 illustrates the use case diagram and the various users' interactions of Student, Admin, and the Assignatory. In this diagram it provides the flow on how the LITES will work.

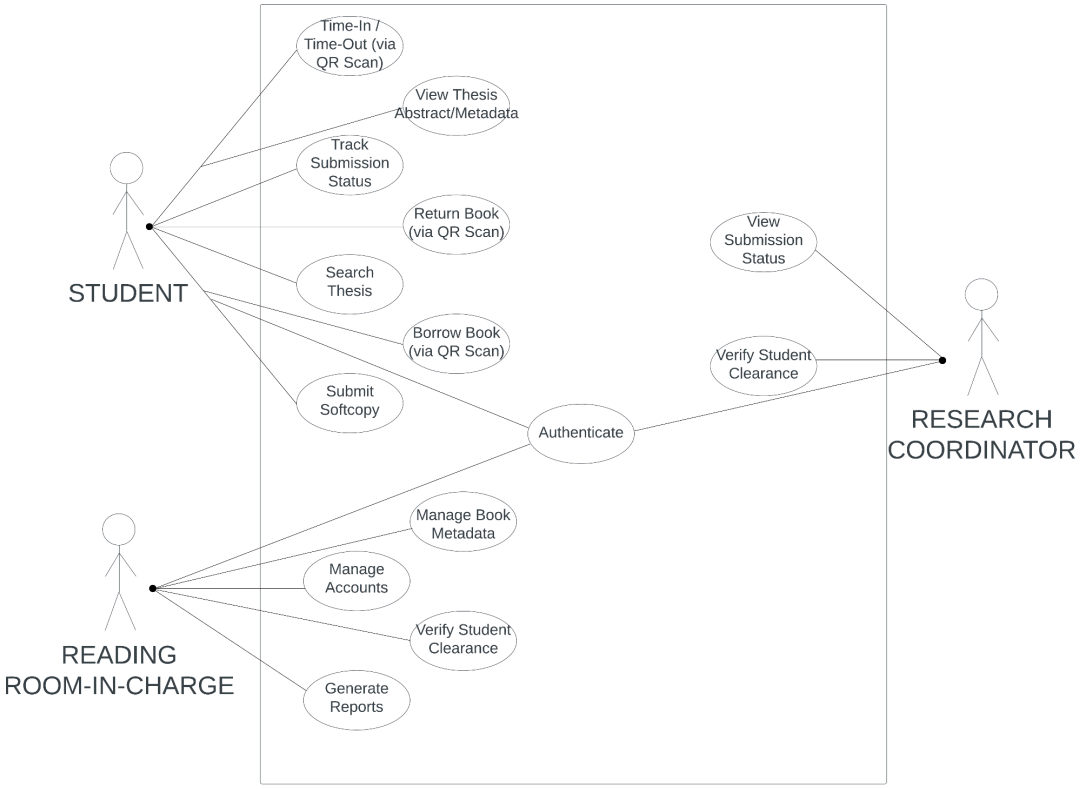

Figure 3. Use Case Diagram of Student, Admin (Reading Room-in-Charge), and Assignatory (Research Coordinator) of LITES

Low-level Data Flow Diagram

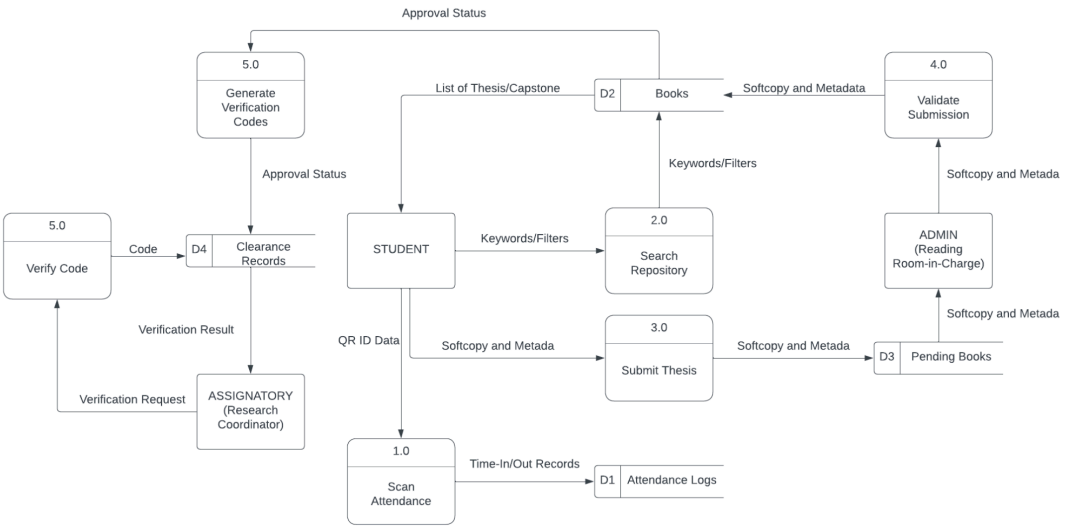

Figure 4. Low-level Data Flow Diagram of LITES: Library of IT and Engineering Studies for CEIT Reading Room

The diagram illustrates the process of archiving a thesis and monitoring student activity within the CEIT Reading Room. The flow begins with the student providing their QR ID Data to Process 1.0, Scan Attendance, which logs the time-in/out record into the Attendance Logs storage (D1). When a student needs to find research, they send Keywords/Filters to Process 2.0, Search Repository, which pulls the List of Thesis/Capstone from the Books storage (D2) and sends the results back. For submission, the student provides Softcopy and Metadata to Process 3.0, Submit Thesis, which places the files into the Pending Books storage (D3). The Admin then performs Process 4.0, Validate Submission, moving approved records into the main Books storage (D2). The final workflow involves the Assignatory (Research Coordinator): the Admin's approval status triggers Process 5.0, Generate Verification Codes, which stores a unique code in Clearance Records (D4). The Assignatory uses this to verify the student via the Verify Code process before providing a final Verification Result for clearance signing.

In this study, students should have the power to register, administer their records, and perform various library actions such as searching, QR attendance, and submitting softcopies. Reading Room-in-Charge (Admin) should have control over account confirmation and book validation. The Research Coordinator (Assignatory) should have the power to verify final thesis submissions before physical clearance signing within the necessary context. The LITES system implements these activities, which generate unique verification codes to invoke the clearance confirmation. The Admin is the primary manager inside the context of the scenario. The Admin can administer user accounts, create reports, and manage the research details. All the users interact with each other using several use case functionalities depending on the interaction type, ensuring that LITES meets all the requirements for a modern, secure, and efficient digital repository.

Activity Diagram

The process begins with the STUDENT selecting "Start" to initiate interaction with the LITES system. The system first prompts them to Display Login Form. The student inputs their email and password, which the system validates (Valid Input?). If the input is invalid ("NO"), the student is prompted to input their credentials again.

If the input is valid ("YES"), the system displays the Dashboard. The student can then select "Add Thesis/Capstone" and Generate Request. This action sends a notification to the ADMIN who will then Review Submission.

At the review stage, the Admin decides if the submission is approved ("YES") or not ("NO").

- If "NO", the student is sent back to the Display Dashboard state to adjust their submission.
- If "YES", the Admin proceeds to Save Book in Database and the system Generate Verification Code. This is the non-decipherable code used for the secure clearance process.

After the code is generated, the student is notified to View Verification Code through their dashboard. The ADMIN and the ASSIGNATORY are also notified to View Verification Code within their respective system modules.

The final stage of the process involves the clearance:

- The Assignatory (Research Coordinator) and the Student interact to Match Code within the system.
- Once the code is matched and verified, the Assignatory can proceed to Sign Clearance, concluding the process.

Figure 4. illustrates the activity diagram of LITES: Library of IT and Engineering Studies for the CEIT Reading Room. It provides the flow of the transaction starting from the login process through thesis submission, approval, and the final clearance verification process.

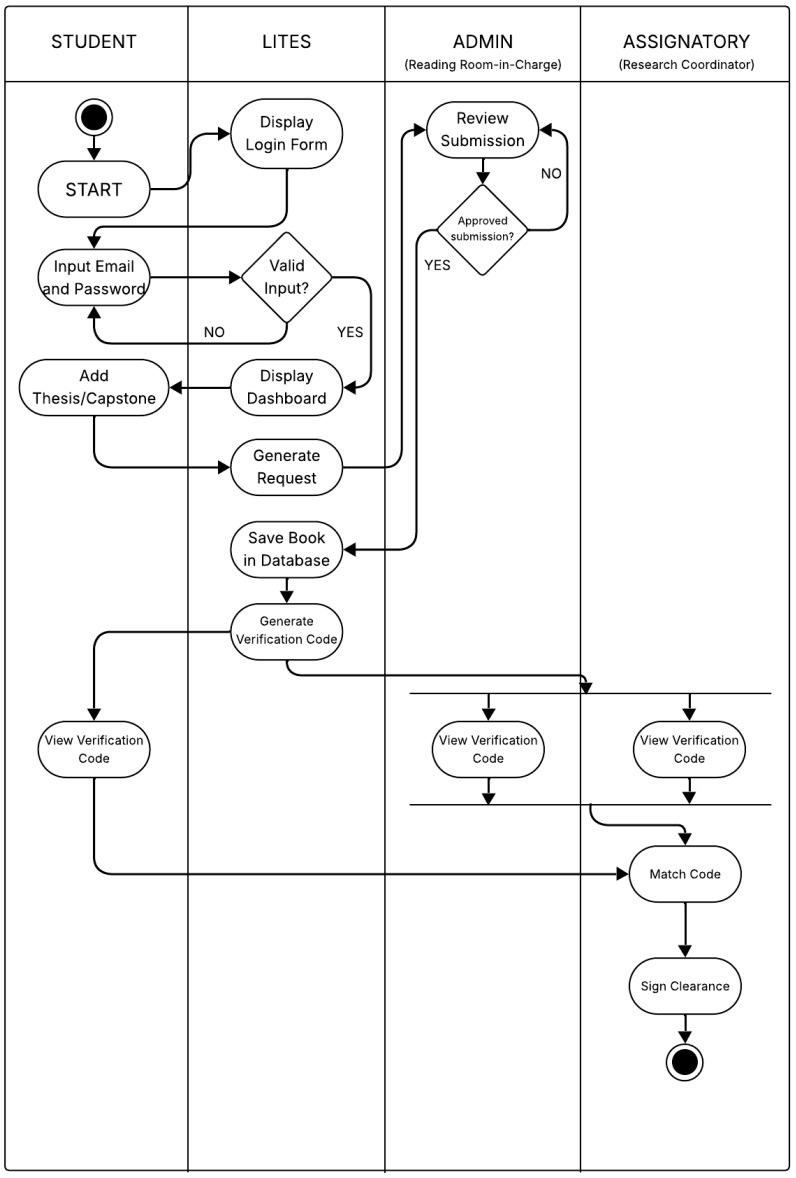

Figure 5. Activity Diagram of LITES: Library of IT and Engineering Studies for CEIT Reading Room

System Design

The development of LITES (Library of IT and Engineering Studies) requires a systematic approach to fulfill the outlined functional and non-functional requirements. To ensure efficient data management of research records, MySQL will be employed as the database system. This system will ensure proper organization and easy retrieval of student information, attendance logs, and thesis metadata, including abstract and submission status data. For ease of use by both administrators and students, system design for interfaces is created with React.js, which provides visual representations of various functionalities from user registration and login processes up to profile management and the online thesis submission portal. To meet security issues related to intellectual property, strong data safety measures will be introduced. This includes the control of accessibility, encryption strategies for sensitive data, and secure ways for logging in (e.g., using CEIT credentials). These methods are taken to guarantee the visibility of the documented security measures and that they follow CEIT's security framework.

In the thesis submission and archiving process, the researchers will develop user-friendly interfaces using React for web-based access, enabling students to access upload features conveniently. The system will collect necessary thesis information like title, abstract, authors, and technical stacks, while also facilitating multi-level approval workflows based on submission status. Additionally, the system will support the upload and verification of required documents, such as the softcopy (PDF) and proof of university affiliation, using backend functionalities provided by XAMPP. To automate library workflows, we will design an intuitive dashboard interface where students can initiate and track their project submissions. Decision support tools will be integrated to assist administrators in objectively evaluating

thesis requests, ensuring efficient and streamlined processing of research archiving. Backend processes will be handled by XAMPP for data automation, allowing for faster and smoother record-keeping within the system.

Technically, JavaScript has been selected as the core for developing the system due to its flexibility and efficacy, particularly with the React framework. The system can be reached using any standard web browser which ensures consistency and simplicity in its operation (adhering to Usability requirements). User activity logs and clear documentation will be included in the system to address training and support needs. The technical support mechanisms, upon establishment, will take care of any issues that may arise without delay. This is done for one single purpose: to ensure the success and efficiency of LITES: Library of IT and Engineering Studies for the CEIT Reading Room.

Login Page for Administrator

Figure 6. illustrates the login page in a web-based of the LITES. Provides secure access for registered users to their accounts especially for Admin(Reading Room-in-Charge) and Assignatory (Research Coordinator). Admin and Assignatory accounts are required to use their accounts and their preferred password for logging in. Upon accessing the login page, registered users input their account details to securely access their accounts within the loan management system.

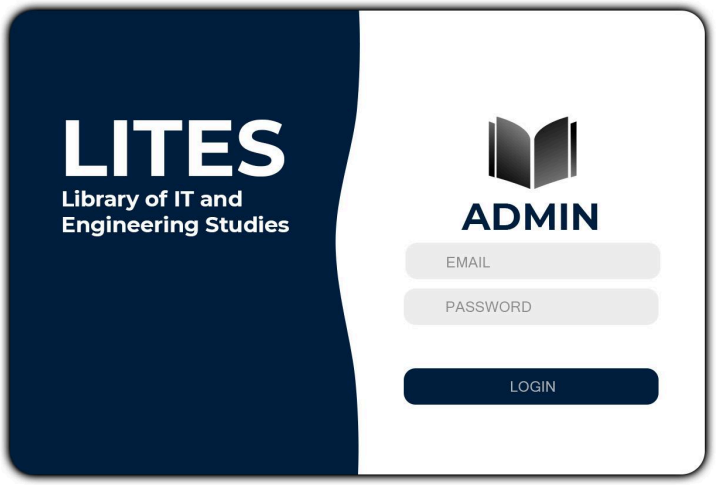

Figure 6. Web-Based Student Login Page of LITES for CEIT Reading Room.

Login Page for Students

Figure 7. illustrates the login page in the web-based of the LITES, the students will be allowed to login their accounts when it is officially created by the admin.

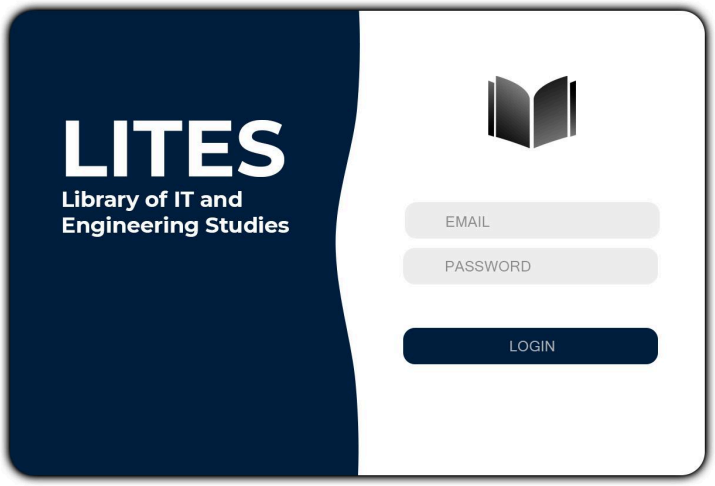

Figure 7. Web-Based Admin Login Page of LITES for CEIT Reading Room.

CEIT Reading Room Logs with QR and 2D Scanner

Figure 8. illustrates the record of a student's attendance and activity within the CEIT Reading Room. It includes details such as the student's ID, name, program, time of entry, time of exit, duration of stay, and attendance remarks ("IN" or "OUT"). The Reading Room Staff (Admin) uses this information to monitor visitor frequency, track student activity patterns, and generate reports on the utilization of

the facility. This system provides vital data for managing the Reading Room's resources effectively.

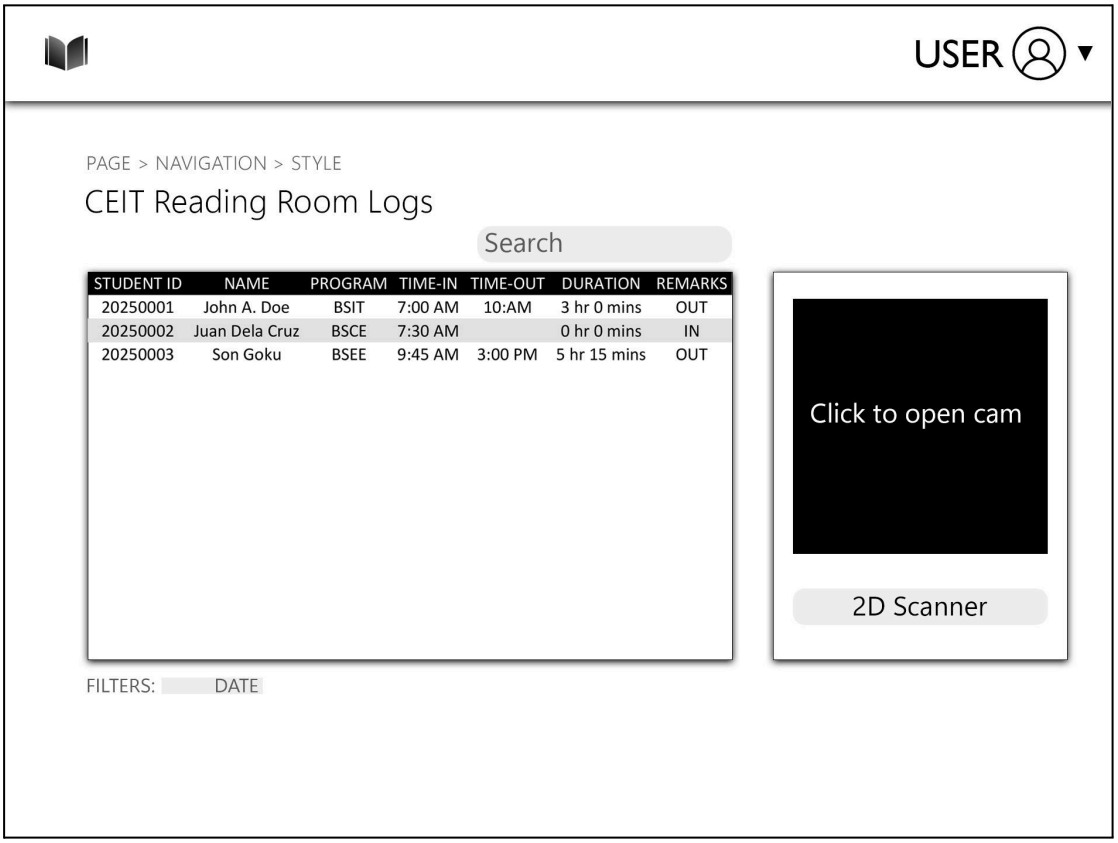

Figure 8. CEIT Reading Room Logs of LITES

Borrowing and Returning of Book Logs

Figure 9. illustrates the record of an individual's borrowing and returning activities within the CEIT Reading Room. The table captures vital information for each transaction, including the unique identification details of the research manuscript (Book ID, Book Name, Book Type) and the borrower (Name of borrower, Email, Program). It also tracks the transaction timeline through the Date Borrowed, Expected Return Date, and the actual Date Returned, with clear Remarks indicating the book's current status (e.g., "BORROWED Due 1 Day").

The LITES system utilizes a two-stage scanning process to manage these transactions efficiently. Stage 1 involves the ID scanning of the student using their unique LITES QR Code, which automatically populates their details in the log. Stage 2 involves the ID scanning of the book, using the distinct QR Code assigned to the physical manuscript. This automated, two-stage workflow ensures accuracy in tracking the movement of limited research resources, providing the Reading Room Staff (Admin) with real-time data to manage the physical inventory and identify overdue items, thereby closing the accountability gap in resource management.

USER

PAGE > NAVIGATION > STYLE

BOOK LOGS

Search

BOOK ID	BOOK NAME	BOOK TYPE	NAME OF BORROWER	EMAIL
CS-IT-0001	Introduction to Computing	Capstone	Juan Dela Cruz	juan.delacruz@example.com
CS-IT-0002	Javascript for Beginners	Thesis	John A. Doe	john.doe@example.com

Click to open cam

2D Scanner

FILTERS: DATE

USER

PAGE > NAVIGATION > STYLE

BOOK LOGS

Search

PROGRAM	DATE BORROWED	DATE EXPECTED RETURN	DATE RETURNED	REMARKS
BSCE	1-Jan-00	2-Jan-00	2-Jan-00	RETURNED (ON-TIME)
BSIT	1-Jan-00	2-Jan-00		BORROWED (DUE 1 DAY)

Click to open cam

2D Scanner

FILTERS: DATE

Figure 9. Book Logs wireframe for Borrowing and Returning of books for CEIT Reading Room

Student Dashboard with Book Searching

Figure 10. illustrates the dynamic research searching interface within the LITES (Library of IT and Engineering Studies) student user interface. This screen is the primary tool for students to locate relevant research manuscripts stored in the digital repository. The UI features a main search bar complemented by advanced filters, allowing users to narrow results by specific criteria such as program (BSIT, BSCE), publication year, or relevant keywords and technical stacks (e.g., React, Arduino).

The interface displays search results in an easily scannable list format. When a student selects a title, the system presents the full metadata, including the abstract, author names, and the technologies used in the project. This functionality is a key component of LITES, designed to efficiently guide students through the literature review process, help them assess the scope of existing studies, and ultimately prevent the duplication of research topics within the CEIT college.

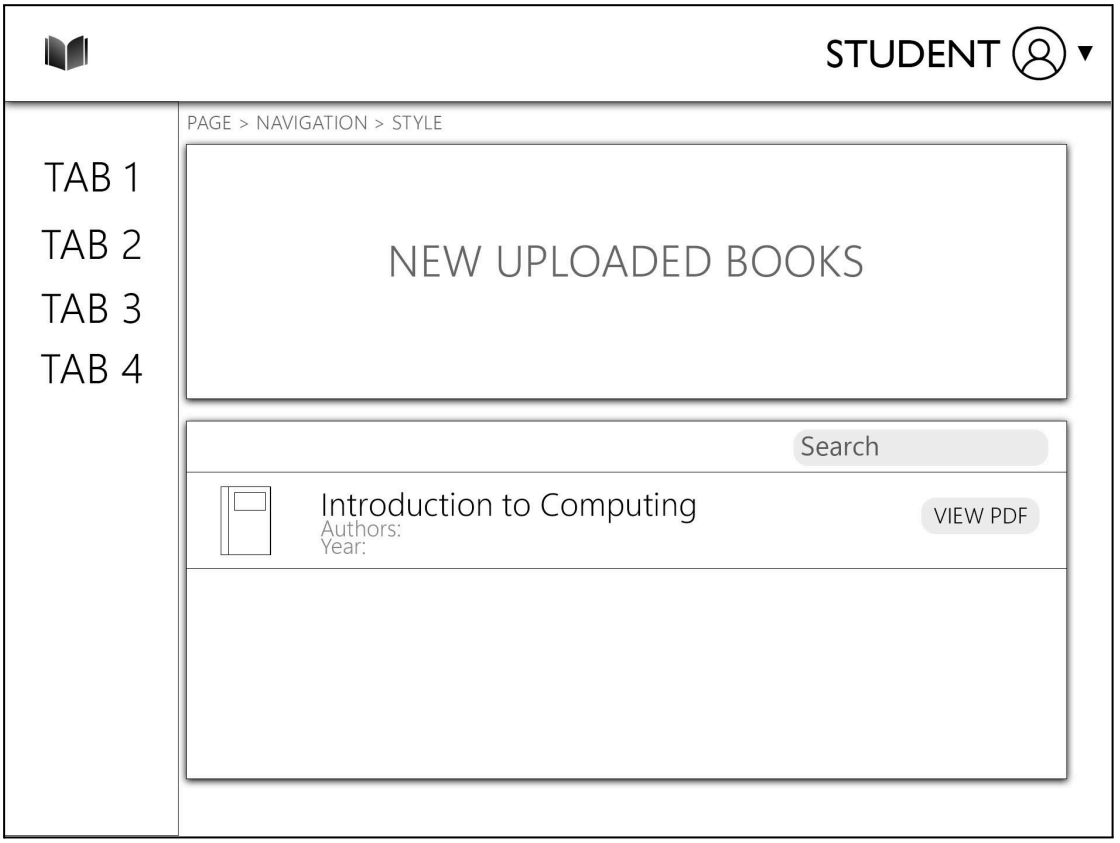

Figure 10. Student Dashboard Wireframe of LITES for CEIT Reading Room

Software Architecture

Figure 11. illustrates the software architecture for LITES: Library of IT and Engineering Studies. The design facilitates efficient CEIT Reading Room operations, focusing on research archiving, student monitoring via QR codes, and reporting functionalities. The architecture is modular, with different parts working together smoothly to help users manage library resources and access important information easily.

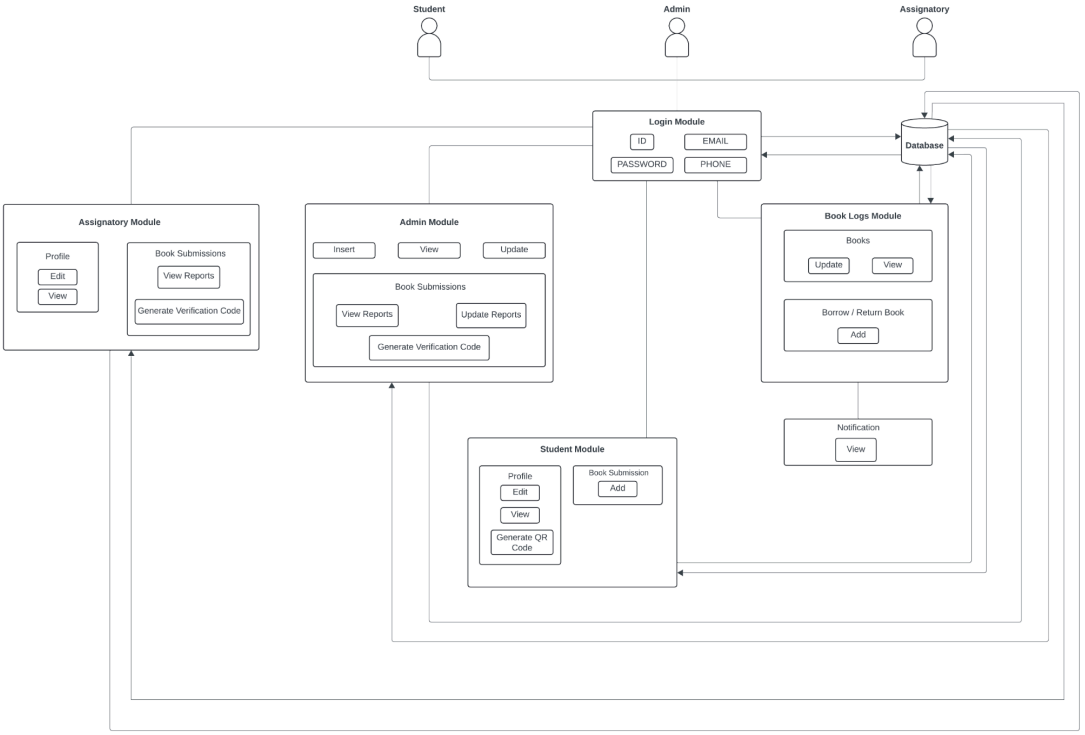

Figure 11. Software Architecture of LITES for CEIT Reading Room

Login Module

The Login module is responsible for authenticating authorized CEIT students, staff, and faculty, granting secure access to the system. As shown in the diagram, users provide credentials (ID, EMAIL, PASSWORD, PHONE). This process involves verifying the data against the centralized Database. Upon successful authentication, the system establishes a session based on the user's role (Student, Admin, or Assignatory), ensuring access to personalized functionalities and maintaining the security framework of the university. The module includes standard security measures like password encryption to enhance data protection.

Admin Module

The Admin module provides the library staff capabilities for managing the repository. This is achieved through several specialized components:

- Book Submissions: Allows the administrator to Insert, View, and Update the status of pending student thesis uploads.
- View Reports / Update Reports: The system supports various reports (Report Generation) related to attendance logs and submission counts, interfacing with the underlying database to present data in a user-friendly format.
- Generate Verification Code: This critical feature allows the admin to trigger the system to create the unique, non-decipherable code used for clearance verification by the Assignatory.

Student Module

The Student module provides essential functionalities for CEIT students (researchers). This module allows students to:

- Manage Profiles: View and Edit personal information.
- QR-Based Attendance: The Exit and View functions are linked to the automated attendance tracking via QR scanning.
- Book Submission: The Add function links to the upload portal for softcopies and metadata.
- Generate QR Code: Students can generate and print their unique QR ID for attendance and borrowing purposes.

Overall, the Student module enhances self-service capabilities and contributes to a transparent and efficient research management system at the CEIT Reading Room.

Assignatory Module

This module is designed for the Research Coordinator, who is responsible for final clearance approval. It includes:

- Profile: Allows the Assignatory to manage their credentials.
- Book Submissions / View: Provides a restricted view of student submissions that require final verification.
- Generate Verification Code: This function is used by the coordinator to input the student's unique code and verify submission status against the database before signing clearances.

Book Logs Module

This module manages physical inventory tracking using QR codes. It includes functions to Update and View book statuses, and the Add functionality specifically manages the Borrow / Return Book transactions, interfacing directly with the database to ensure data integrity and track overdue items.

All modules within LITES are tightly integrated with a centralized Database. This database serves as the foundation for storing and managing critical information related to research records, user profiles, transaction histories, attendance logs, and notifications. Each module interacts with the database via the data flows shown in Figure 10, ensuring data integrity, system reliability, and overall performance across the LITES platform.

System Development

In developing the proposal for LITES: Library of IT and Engineering Studies for the CEIT Reading Room, an Iterative Process Methodology will be employed for

the system. System development is governed by the Iterative process development life cycle, which consists of 7 phases: Initialization, Planning, Requirements, Design, Implementation, Verification, and Deployment. These phases serve as the framework for implementing the specialized features of LITES, ensuring the system fulfills the specific archiving and monitoring needs of the college.

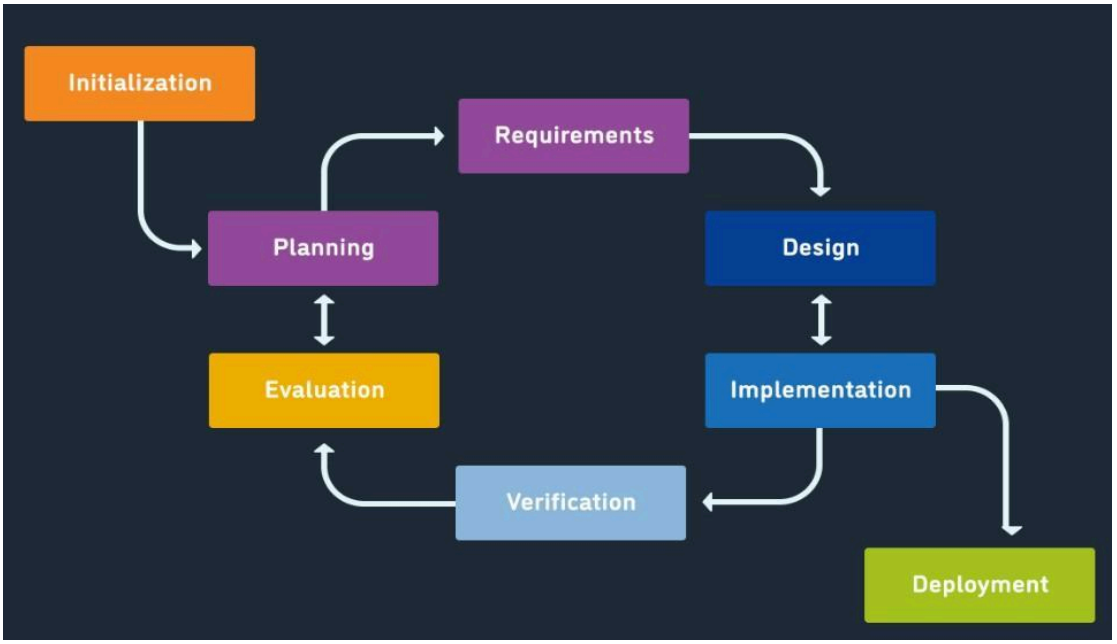

Figure 12. Iterative process development life cycle (Trunkett, 2020)

The Initialization Phase marks the beginning of the project to develop LITES for the CEIT Reading Room. During this phase, the project objectives—such as digitalizing research archives and automating attendance—are defined alongside the scope and feasibility of the system. Stakeholders, including the Reading Room-in-Charge, Mr. Ace Amiel E. Malicsi, and the Research Coordinator, are identified and engaged to ensure alignment with the college's goals. The process began with an exhaustive Initialization phase, consisting of extensive interviews to elicit specific needs regarding the QR-based entry system and the clearance submission workflow.

In the Planning Phase, the researchers create detailed plans for developing the LITES platform. Resources such as hardware (QR scanners), software (React, MySQL), and technological needs are allocated based on the project requirements. Project timelines, milestones, and goals for each iteration of development are established. Potential risks, such as the encryption of the non-decipherable coordinator code, are identified with corresponding management strategies. Each iteration is carefully planned to define specific deliverables, ensuring a well-organized approach to building the library repository.

Moreover, in the Requirements Phase, the researchers gather and document all the needs and expectations for the LITES system. Consultations are held with the staff to understand specific functionalities, such as the dynamic searching of technical stacks and the automated "Time-In/Time-Out" features. These requirements are prioritized based on their importance to the Reading Room's daily operations. Throughout the development process, the requirements are continuously refined and updated to adapt to any changes or new insights that arise from the 300+ seasonal visitor patterns.

The Design Phase involves creating a blueprint for how LITES will function and appear. This involves designing the structure of the system, including the MySQL database schema for organizing student profiles and thesis metadata. The User Interface (UI/UX) is designed with a focus on responsiveness, ensuring that students can easily interact with the system via mobile devices for QR scanning. Feedback from the staff is used to refine these designs, ensuring they meet the high standards expected for the Reading Room renovation.

In the Implementation Phase, the researchers begin building LITES based on the established designs. This involves writing the actual code using JavaScript and React for the front-end and XAMPP/MySQL for the backend. The work is broken

down into manageable tasks, such as coding the QR scanning module and the research filtering logic. As progress is made, components are continuously integrated and tested to ensure they work together smoothly, turning the design into a functional archiving system that aligns with CEIT's requirements.

In the Verification Phase, the researchers thoroughly test LITES to ensure it works correctly and meets all academic requirements. This involves different types of testing, such as unit testing for the QR code generation, integration testing to verify that the softcopy upload correctly triggers the clearance verification code, and User Acceptance Testing (UAT) with Mr. Malicsi to ensure the system meets staff expectations. Any identified bugs are fixed to ensure the system is reliable and secure before deployment.

Lastly, in the Deployment Phase, the completed LITES system is released for actual use by the CEIT Reading Room. This involves installing the system on the college's production servers and configuring it for the local network. Training and support are provided to the library staff and students to help them adapt to the QR-based operations. Continuous monitoring and maintenance are carried out to ensure LITES runs smoothly and efficiently, fulfilling its purpose as a modernized research archive.

Developing LITES follows a structured iterative process that allows for continuous improvement and adaptation. This ensures a successful implementation that enhances accessibility and secures the research legacy of the College of Engineering and Information Technology (CEIT).

System Testing

For the administrators, a series of tests are designed to validate critical functionalities of LITES. The Reading Room-in-Charge will perform the login process, manage student accounts (adding and verifying CEIT-exclusive registration), approve or reject uploaded thesis softcopies, and assign distinct QR codes to physical hardbound books. For the Research Coordinator, testing will focus on the accuracy of the non-decipherable verification code generated by the system to ensure it serves as a secure proof of submission. For the students, testing will encompass all front-end features, specifically the QR-based Time-In/Out, dynamic research searching, and the manuscript upload portal.

Usability testing will assess the overall user interface of LITES, focusing on mobile responsiveness for QR scanning, intuitive navigation through research categories, and consistency in design to address the UI/UX failures of previous systems. The aim is to ensure an intuitive and user-friendly experience for the 300+ seasonal visitors. User experience (UX) testing will evaluate the system's interaction flow, such as the speed of the "Search-to-Abstract" transition. Feedback from CEIT students and staff will be gathered to identify areas for improvement. Additionally, accessibility testing will be conducted to ensure the system is usable for individuals with disabilities, including compatibility with screen readers for viewing digital abstracts.

System Evaluation

The evaluation of LITES: Library of IT and Engineering Studies will primarily focus on its functional suitability and the security of the research archive. The researchers will assess the thesis submission process to ensure it is efficient and user-friendly, allowing students to easily upload documentation and metadata. The efficiency of the archiving workflow will be evaluated, examining the librarian's criteria for approving uploads and the transparency of communication regarding the student's clearance status.

Furthermore, the system's automated attendance process will be assessed to ensure accuracy and timeliness in recording student logs via QR scanners. Additionally, the researchers will examine how the system manages the borrowing and returning of physical books, including tracking due dates and sending automated return reminders. The system's performance will be compared to the initial requirements and the ISO 25010 standards, ensuring every aspect aligns with the college's goal of digital transformation. Through rigorous testing, the researchers will ascertain the system's reliability, aiming for a digital repository that secures the college's research legacy for years to come.

Special attention will be paid to the interface usability, ensuring that the dynamic filtering of technical stacks is as smooth as operating a modern search engine. User feedback will serve as a critical guide for refinements, ensuring the system resonates with the specific academic needs of CEIT. The researchers will also evaluate the effectiveness of the training provided to the Reading Room staff, ensuring they are well-equipped to troubleshoot the system and manage the growing digital database.

System Implementation

The development of LITES will be transferred to the CEIT Reading Room along with detailed documentation that serves as a comprehensive guide for the Reading Room-in-Charge. This documentation will provide a thorough understanding of the system's architecture, the QR generation process, and the MySQL database settings, empowering the administrator to effectively manage updates and maintenance. Through personalized training sessions, the researchers will ensure that Mr. Ace Amiel E. Malicsi and the library staff feel confident in utilizing all features, from attendance monitoring to clearance verification.

The documentation will not only cover basic functionalities but also delve into administrative procedures such as handling "delinquent" submissions and generating research trend reports. It will serve as a living resource that adapts to the evolving needs of the college. The guide will be designed with user-friendliness in mind, featuring clear instructions, database diagrams, and screenshots of the admin dashboard.

In line with our commitment to ongoing support, a dedicated support channel will be established where the CEIT staff can reach out for assistance. In this manner, LITES transforms into a dependable companion for the College of Engineering and Information Technology, reinforcing the department's dedication to modernized research management and academic excellence.

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